

Warmup

3rd YEAR - ENSIA

Exercise C1 (Solving linear system + Gauss-Jordan)

A. Solve the the following linear system using the Gauss-Jordan method and examine the solvability based on the values of b :

$$\begin{cases} x_1 + x_2 = 1 \\ x_2 + x_3 = 3 \\ x_1 + 2x_2 + x_3 = b \end{cases}$$

B. Calculate the inverse of the following matrix using Gauss-Jordan method:

$$\begin{pmatrix} 2 & 0 & 1 \\ 3 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

C. Deduce the solution of the following system:
$$\begin{cases} 2x_1 + x_3 = 5 \\ 3x_1 + x_2 = 15 \\ x_1 + 2x_3 = 10 \end{cases}$$

Exercise C2 (Linear independence)

Determine if each of the following sets of vectors is linearly independent or linearly dependent.

$$V_1 = \{[1 \ 0 \ 1], [1 \ 2 \ 1], [2 \ 2 \ 2]\}$$

$$V_2 = \{[2 \ 1 \ 0], [1 \ 2 \ 0], [3 \ 3 \ 1]\}$$

$$V_3 = \{[2 \ 1], [1 \ 2]\}$$

$$V_4 = \{[2 \ 0], [3 \ 0]\}$$

$$V_5 = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix} \right\}$$

$$V_6 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Exercise C3 (Modeling with linear system)

Only three brands of juice (juice 1, juice 2, and juice 3) are available for sale in a cafeteria. From time to time, people try one or another of these brands. Suppose that at the beginning of each month, people change the juice they are drinking according to the following rules:

- 30% of the people who prefer juice 1 switch to juice 2.
- 20% of the people who prefer juice 1 switch to juice 3.
- 30% of the people who prefer juice 2 switch to juice 3.
- 30% of the people who prefer juice 3 switch to juice 2.
- 10% of the people who prefer juice 3 switch to juice 1.

For $i = 1, 2, 3$, let x_i be the number who prefer juice i at the beginning of this month and y_i be the number who prefer juice i at the beginning of next month. Use matrix multiplication to relate the following vectors: $\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Exercise P1 (Optional)

Suppose that a linear system $Ax = b$ has more variables than equations. Show that $Ax = b$ cannot have a unique solution.

Exercise P2 (Optional)

Show that a square matrix has an inverse if and only if its rows form a linearly independent set of vectors.

Exercise P3 (Optional)

Before paying employee bonuses and state and federal taxes, a company earns profits of \$60,000. The company pays employees a bonus equal to 5% of after-tax profits. State tax is 5% of profits (after bonuses are paid). Finally, federal tax is 40% of profits (after bonuses and state tax are paid). Determine a linear equation system to find the amounts paid in bonuses, state tax, and federal tax.

Exercise P4 (Optional)

Each year, 20% of all untenured State University faculty become tenured, 5% quit, and 75% remain untenured. Each year, 90% of all tenured S.U. faculty remain tenured and 10% quit. Let U_t be the number of untenured S.U. faculty at the beginning of year t , and T_t the tenured number. Use matrix multiplication to relate the two vectors $\begin{bmatrix} U_{t+1} \\ T_{t+1} \end{bmatrix}$ $\begin{bmatrix} U_t \\ T_t \end{bmatrix}$

Exercise P5 (Optional)

The grades of two students during their last semester are shown in Table 1. Courses 1 and 2 are four-credit courses, and courses 3 and 4 are three-credit courses. Let GPA_i be the semester grade point average for the student i . Use matrix multiplication to express the vector $\begin{bmatrix} GPA_1 \\ GPA_2 \end{bmatrix}$ in terms of the information given in the problem.

Table 1

Student	Course			
	1	2	3	4
1	3.6	3.8	2.6	3.4
2	2.8	3.1	2.9	3.6